

Statistical Modeling

DATA.STAT.740

Lecture 2

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Normal linear model: single binary predictor

Let the predictor have two possible values $x_i \in \{A_1, A_2\}$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Normal linear model is now a set of normal distributions

$$x_i = A_1 : y_i = \beta_0 + \beta_1^1 + \epsilon_i,$$

$$x_i = A_2 : y_i = \beta_0 + \beta_1^2 + \epsilon_i$$

We can simplify the model a little bit.

Normal linear model: single binary predictor

Let the predictor have two possible values $x_i \in \{A_1, A_2\}$

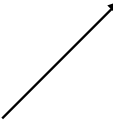
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Normal linear model is now a set of normal distributions

Let's use indicator or dummy coded variables by selecting $x_i = A_1$ as the reference and introducing

$$x_i^{A_2} = \begin{cases} 0, & \text{if } x_i = A_1 \\ 1, & \text{if } x_i = A_2 \end{cases}$$

is $x_i = A_2$



Normal linear model: single binary predictor

Using the dummy variable, the model becomes

$$y_i = \beta_0 + \beta_1^{A_2} x_i^{A_2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

or

$$y_i = \begin{cases} \beta_0, & \text{if } x_i = A_1 \\ \beta_0 + \beta_1^{A_2}, & \text{if } x_i = A_2 \end{cases}, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

where β_0 is the common intercept for both groups and $\beta_1^{A_2}$ is the difference between the group means!

Model is used in the **independent samples t-test**

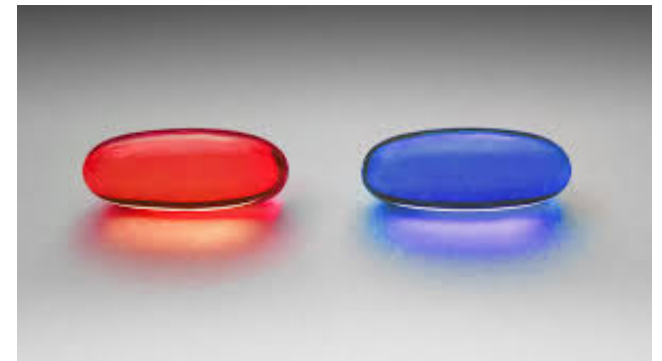
Example: Placebo vs Treatment

Clinical trial: A new drug has been developed to treat disease Z. We wish to test its effectiveness in treating disease Z by eliminating the effect of chance in the treatment outcomes.

Two groups: $2N$ patients with disease Z have been randomly allocated one of the groups.

N patients in group A will receive a sugar pill

N patients in group B will receive a dose of new drug



Normal linear model: single categorical predictor

Let the predictor have K possible values $x_i \in \{A_1, \dots, A_K\}$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Dummy coded variables

is $x_i = A_2$
↙

	$x_i^{A_1}$	$x_i^{A_2}$	$\dots$	$x_i^{A_K}$
$x_i = A_1$	1	0	$\dots$	0
$x_i = A_2$	0	1	$\dots$	0
$\vdots$				
$x_i = A_K$	0	0	$\dots$	1

Normal linear model: single categorical predictor

Dummy coded variables

	$x_i^{A_1}$	$x_i^{A_2}$	$\dots$	$x_i^{A_K}$
$x_i = A_1$	1	0	$\dots$	0
$x_i = A_2$	0	1	$\dots$	0
$\vdots$				
$x_i = A_K$	0	0	$\dots$	1

Exactly one of the dummy variables is always 1 the rest is 0

If we have $K - 1$ dummy variables that are 0, we know which of the variables is 1

We select one of the groups as **the reference Group** (e.g. A_1)

$$y_i = \beta_0 + \beta_1^{A_2} x_i^{A_2} + \dots + \beta_1^{A_K} x_i^{A_K} + \epsilon_i$$

Normal linear model: single categorical predictor

The models for each group:

$$\begin{aligned}x_i = A_1 : \quad y_i &= \beta_0 + \epsilon_i, \\x_i = A_2 : \quad y_i &= \beta_0 + \beta_1^{A_2} + \epsilon_i \\&\vdots \\x_i = A_K : \quad y_i &= \beta_0 + \beta_1^{A_K} + \epsilon_i,\end{aligned}$$

Group models have a common intercept and a coefficient indicating the difference of the average response with respect to the reference group average response

Model is used in **one-way analysis of variance**

Linear regression with a single categorical predictor

The recoded linear normal model is $y = X\beta + \epsilon$ where

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{matrix} A_1 \\ \\ A_2 \\ \\ A_3 \\ \\ A_K \end{matrix}$$

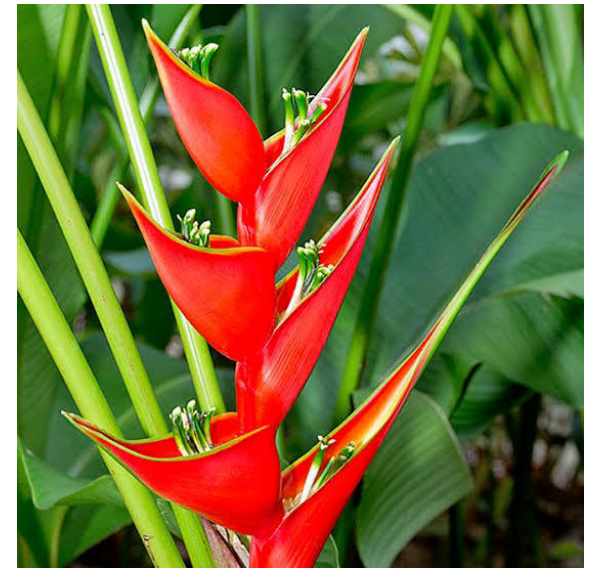
Example: group comparisons

This dataset (heliconia.txt) includes the average lengths of different varieties of heliconia plants

Bihai

Red Caribaea

Yellow Caribaea



Normal linear model: two continuous predictors

Let's investigate the linear normal model with two predictors

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

Let both predictors be continuous

β_0 : intercept

β_i : partial regression coefficients - the change in the outcome mean when the corresponding predictor is changed by one unit and all the others are kept constant

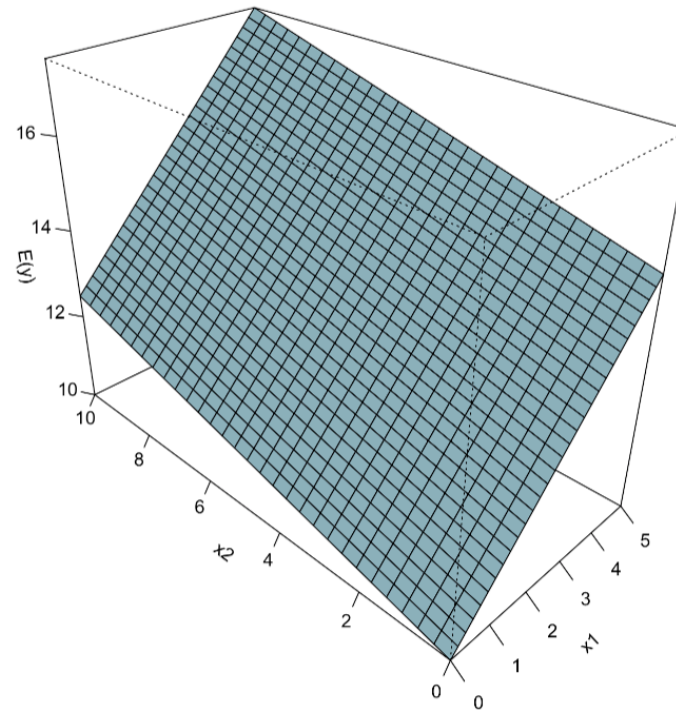
Normal linear model: two continuous predictors

The mean of the model

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

in three-dimensional space is a plane

Let $\beta_0 = 10, \beta_1 = 1, \beta_2 = 0.25$



Normal linear model: two continuous predictors

The mean of the model

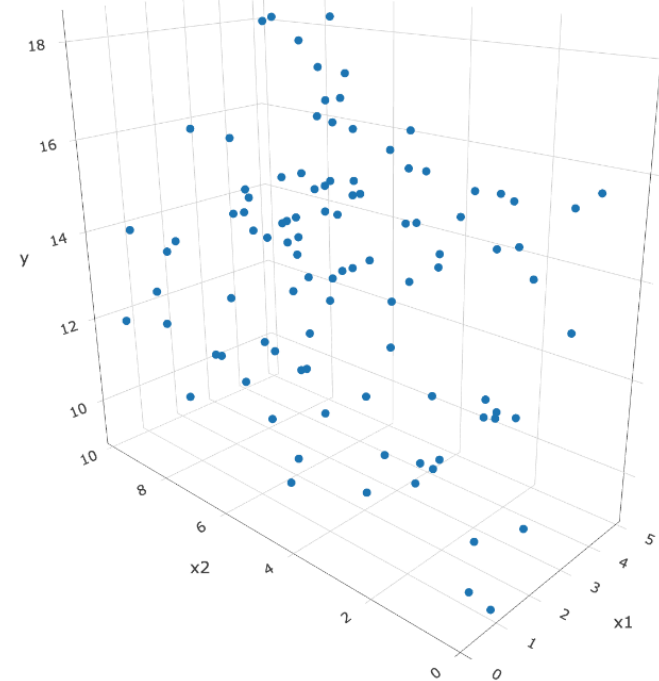
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Let $\beta_0 = 10, \beta_1 = 1, \beta_2 = 0.25$

Add error with $\sigma^2 = 0.2^2$

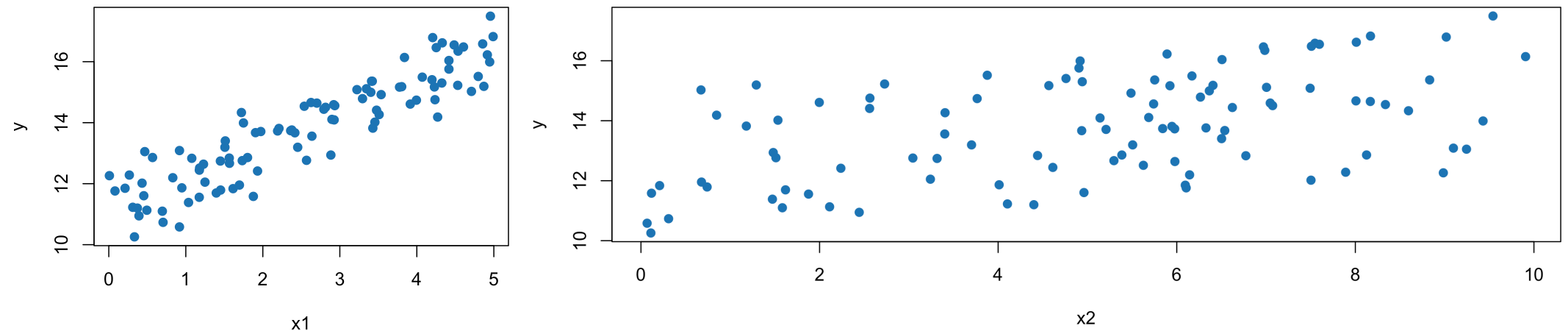
Static 3D-scatterplots of observations
become messy



Normal linear model: two continuous predictors

Investigate data in two dimensional scatter plots:

How much effect? How much error/noise?

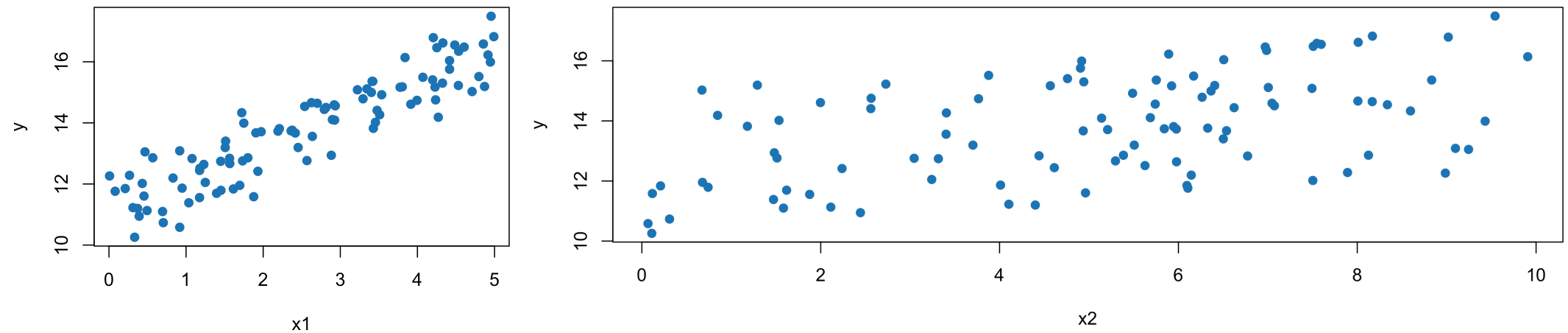


Plots have the same observations

Normal linear model: two continuous predictors

Investigate data in two dimensional scatter plots:

How much effect? How much error/noise?



Plots have the same observations

R-example

Example: Volume of black cherry tree timber

This dataset (blackcherry.txt) includes the measurements of the girth, height and volume of timber in felled black cherry trees

Girth is the diameter of the tree (in inches) measured at 4ft6in above the ground, height is measured in ft and volume of timber is measured in cubic ft

Now, we'll use girth and height as the predictors of the volume of timber



Normal linear model: Two continuous predictors with an interaction

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

Diagram illustrating the components of the model equation:

- Main effects: $\beta_1 x_{i,1}$ and $\beta_2 x_{i,2}$
- Interaction effect: $\beta_3 x_{i,1} x_{i,2}$

The effect of $x_{i,1}$ is not constant - it depends on the value of $x_{i,2}$

$$\frac{\partial}{\partial x_{i,1}} [\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2}] = \beta_1 + \beta_3 x_{i,2}$$

The variable **moderates** the effect of the other

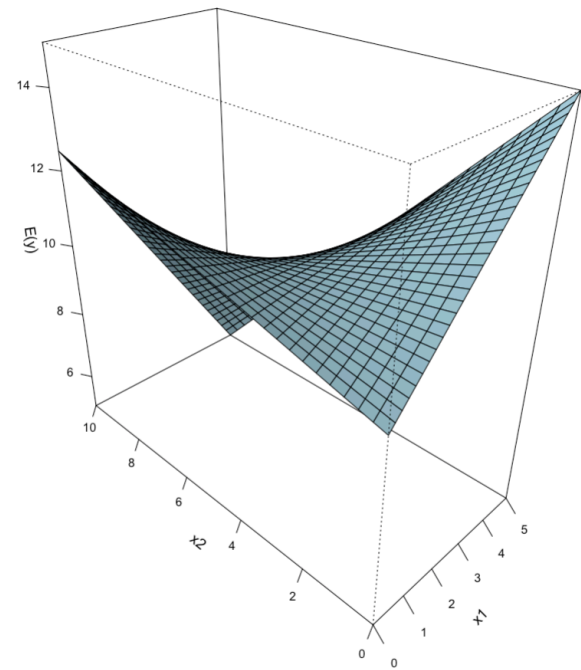
Normal linear model: two continuous predictors

The mean of the model

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in three-dimensional space can be a saddle-like surface

Let $\beta_0 = 10, \beta_1 = 1, \beta_2 = 0.25, \beta_3 = -0.25$



Normal linear model: two continuous predictors

The mean of the model

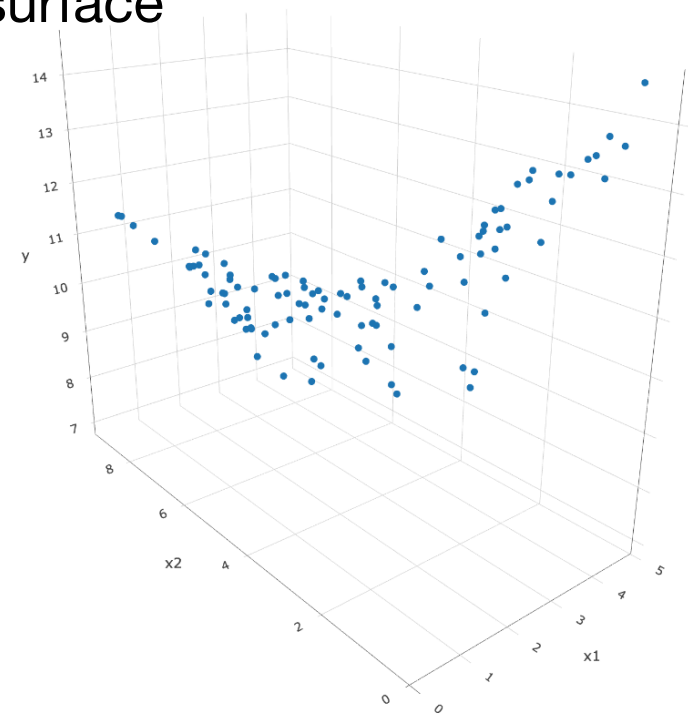
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Add error with $\sigma^2 = 0.2^2$

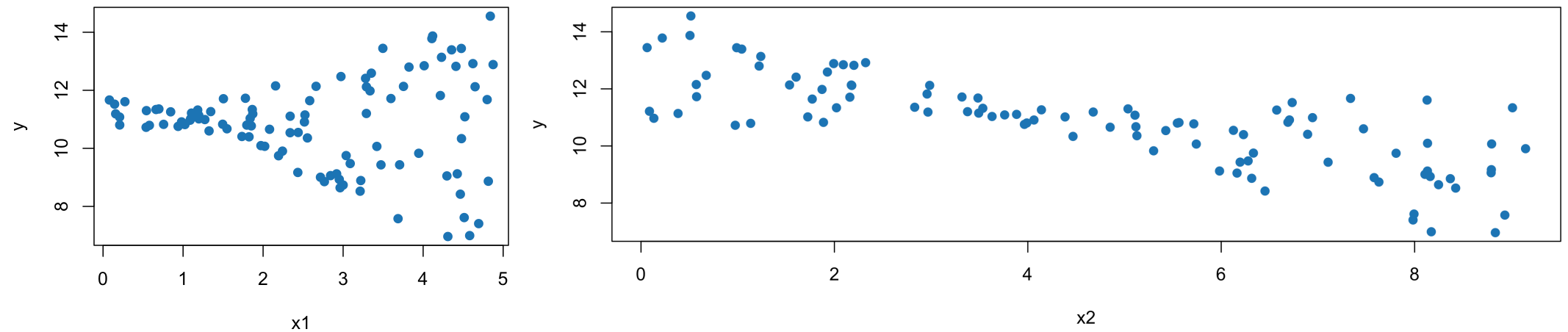
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Normal linear model: two continuous predictors

Investigate data in two dimensional scatter plots

How much effect? How much error/noise?

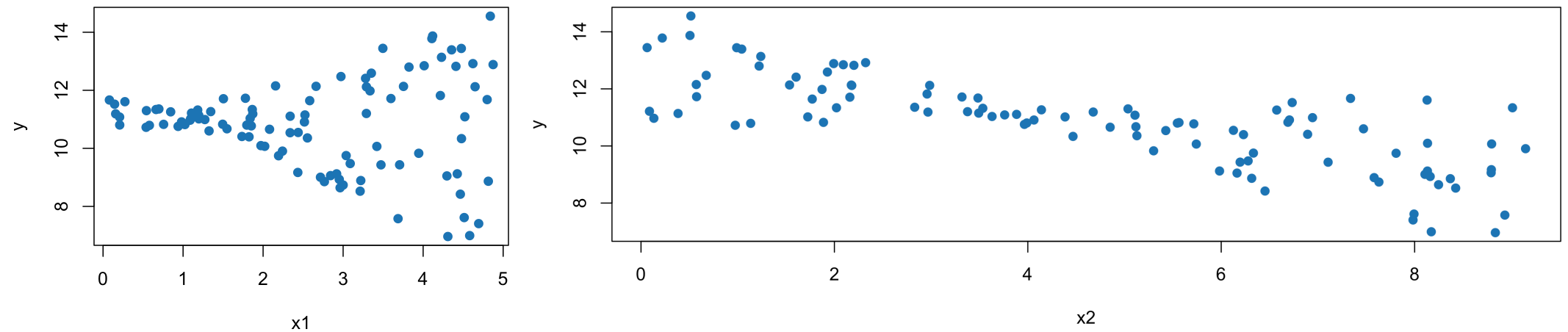


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Normal linear model: two continuous predictors

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Now, we'll use girth and height and their interaction as the predictors of the volume of timber



Two categorical predictors

Two-way ANOVA (without interaction)

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

categorical

$$x_{i,1} \in \{A_1, \dots, A_K\}$$

$$x_{i,2} \in \{B_1, \dots, B_M\}$$

Dummy variables

$$y_i = \beta_0 + \beta_1^{A_2} x_{i,1}^{A_2} + \dots + \beta_1^{A_K} x_{i,1}^{A_K} + \beta_2^{B_2} x_{i,2}^{B_2} + \dots + \beta_2^{B_M} x_{i,2}^{B_M} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

Reference: $x_{i,1} = A_1, x_{i,2} = B_1$

Two categorical predictors

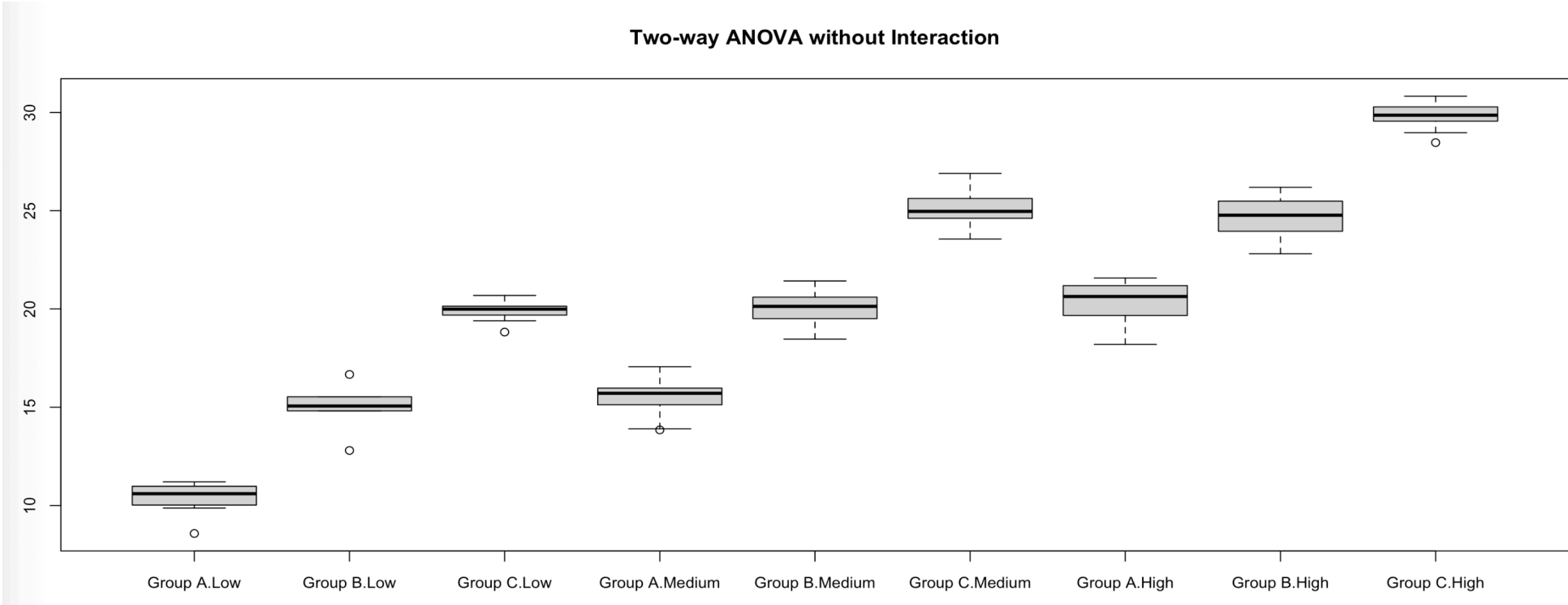
Two-way ANOVA

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

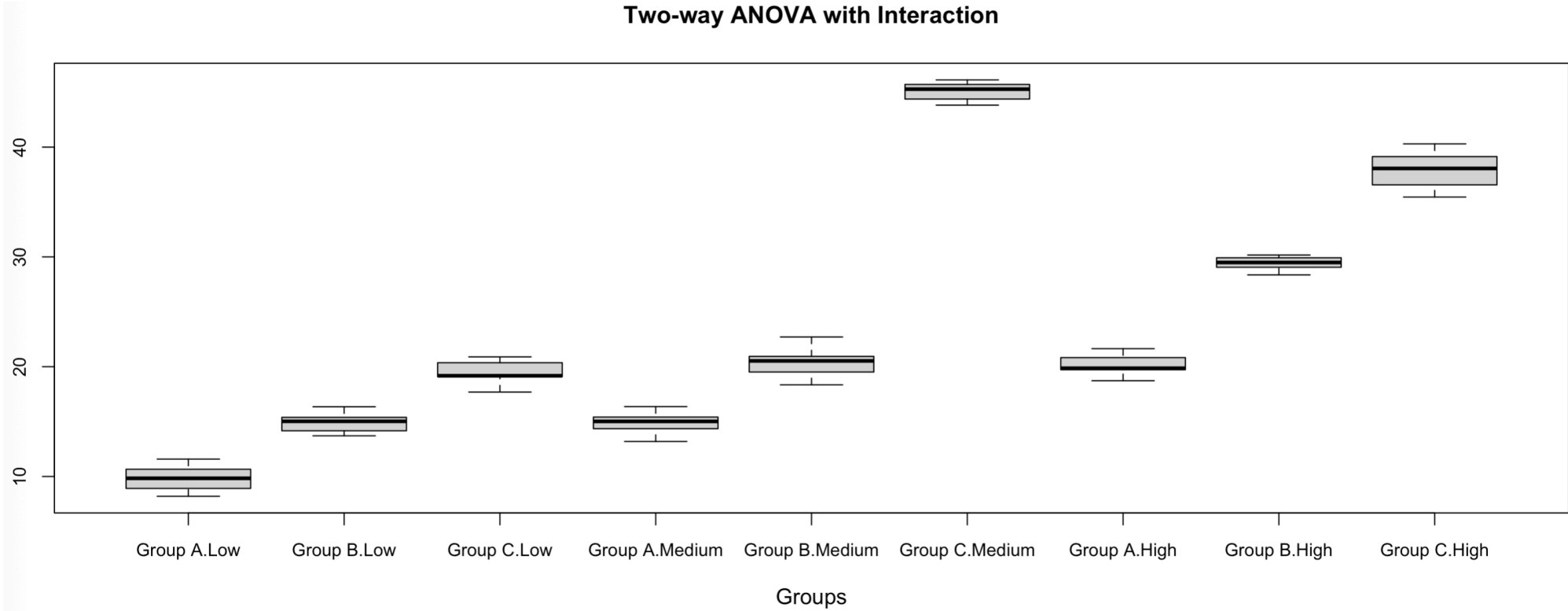
With dummy variables

$$\begin{aligned} y_i = & \beta_0 + \beta_1^{A_2} x_{i,1}^{A_2} + \cdots + \beta_1^{A_K} x_{i,1}^{A_K} \\ & + \beta_2^{B_2} x_{i,2}^{B_2} + \cdots + \beta_2^{B_M} x_{i,2}^{B_M} \\ & + \beta_3^{A_2, B_2} x_{i,1}^{A_2} x_{i,2}^{B_2} + \cdots + \beta_3^{A_2, B_M} x_{i,1}^{A_2} x_{i,2}^{B_M} \\ & \vdots \\ & + \beta_3^{A_K, B_2} x_{i,1}^{A_K} x_{i,2}^{B_2} + \cdots + \beta_3^{A_K, B_M} x_{i,1}^{A_K} x_{i,2}^{B_M} \\ & + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2), \end{aligned}$$

Example: Two-way ANOVA (without interaction)



Example: Two-way ANOVA



One continuous and one categorical predictor

Linear analysis of covariance (no interaction)

Some of the explanatory variables are defined on numerical scale and some on categorical scale

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

$x_{i,1} \in \{A_1, \dots, A_K\}$



K models:

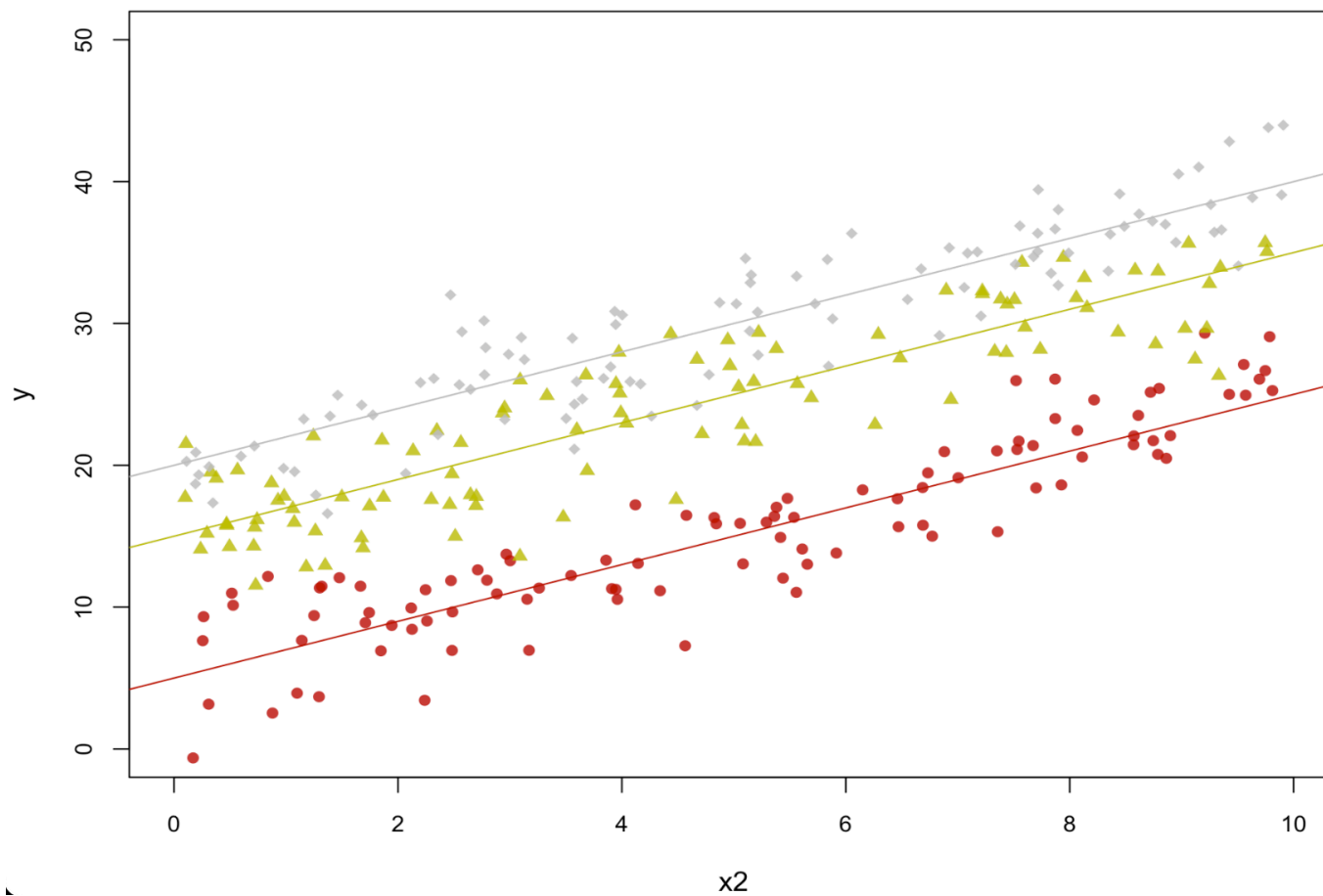
$$x_{i,1} = A_1 : y_i = \beta_0 + \beta_2 x_{i,2} + \epsilon_i,$$

$$x_{i,1} = A_2 : y_i = \beta_0 + \beta_1^{A_2} + \beta_2 x_{i,2} + \epsilon_i$$

$$\vdots$$

$$x_{i,1} = A_K : y_i = \beta_0 + \beta_1^{A_K} + \beta_2 x_{i,2} + \epsilon_i,$$

Example: Linear analysis of covariance (no interaction)



One continuous and one categorical predictor

Linear analysis of covariance (with interaction)

Some of the explanatory variables are defined on numerical scale and some on categorical scale

$$x_{i,1} \in \{A_1, \dots, A_K\}$$
$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \beta_3 x_{i,1} x_{i,2} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2),$$

K models:

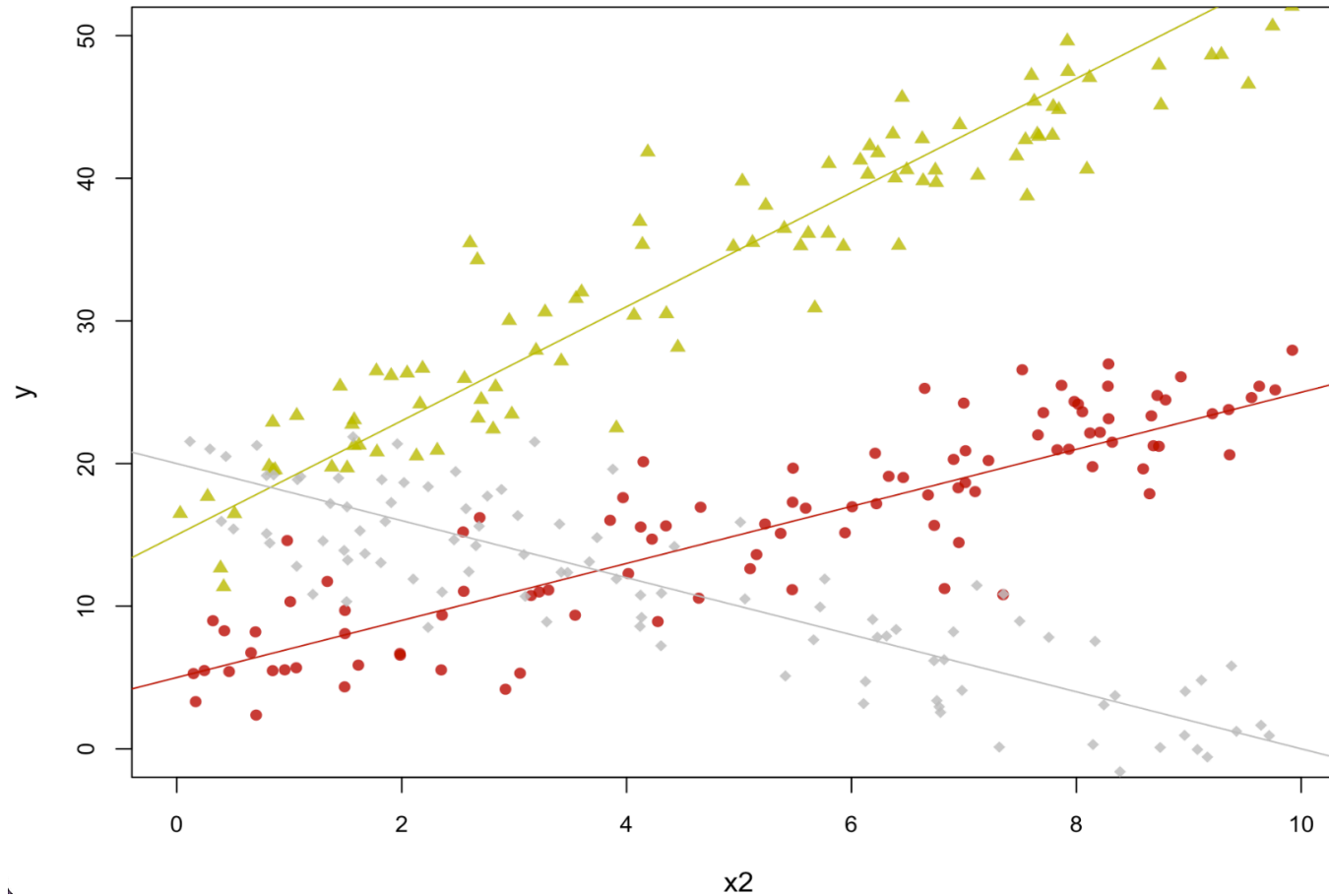
$$x_{i,1} = A_1 : y_i = \beta_0 + \beta_2 x_{i,2} + \epsilon_i,$$

$$x_{i,1} = A_2 : y_i = \beta_0 + \beta_1^{A_2} + \beta_2 x_{i,2} + \beta_3^{A_2} x_{i,2} + \epsilon_i$$

$$\vdots$$

$$x_{i,1} = A_K : y_i = \beta_0 + \beta_1^{A_K} + \beta_2 x_{i,2} + \beta_3^{A_K} x_{i,2} + \epsilon_i,$$

Example: Linear analysis of covariance



End of Lecture 2